

Shubham Sarkar

Machine Learning Engineer

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Detail-oriented Machine Learning Engineer with a strong foundation in designing, training, and deploying scalable ML systems. Proven expertise in optimizing models for performance and cost efficiency, with hands-on experience in MLOps tools and cloud platforms. Adept at collaborating with cross-functional teams to bridge research and production, ensuring high-quality, well-tested code in Python.

Experience

Alignerr

Machine Learning Engineer
Remote

Jan 2026 – Present

- Enhanced LLM evaluation precision by 15% through a comprehensive review of a rubric-based scoring framework across six reasoning categories.
- Analyzed over 50 audio files for integration into ASR pipelines, ensuring high-quality data ingestion.
- Evaluated AI agent responses to identify failure points such as inference memory and self-coherence, leading to a 20% improvement in model accuracy.

Jadavpur University CMATER Lab

Deep Learning Research Assistant
Kolkata, India

May 2025 – September 2025

- Designed and implemented a self-attention mechanism (scaled dot-product) within a pre-trained VGG16, significantly enhancing feature extraction for lung cancer detection from CT scans.
- Developed a hybrid deep learning architecture achieving 99.54% peak accuracy with only 76k trainable parameters and 0.0256 GFLOPs, facilitating edge-device deployment.
- Engineered feature fusion through concatenation and element-wise multiplication of original and attention-modulated maps for refined, context-aware representations.

Education

Jadavpur University, Kolkata

Bachelor of Technology
CGPA: 7.5

Nov 2023 – Dec 2027

Hariyana Vidya Mandir, Kolkata

Higher Secondary
Percentage: 90%

Apr 2020 – Apr 2022

Projects

TailorCV.ai

Python, FastAPI, LLM, AI agents, Amazon Web Services
[Link](#) | [GitHub](#)

2 January 2026

- Developed an AI web application that optimizes resumes to job descriptions using LLMs and NLP pipelines, improving resume relevance by up to 80%.
- Designed a Python and FastAPI backend with HTML, CSS, and JavaScript for the frontend, dockerized the application, and deployed it on AWS ECS.
- Achieved over 50 users within the first week of launch, demonstrating strong early adoption and real-world impact.

YouTube Sentiment Analysis

TensorFlow, NLP, AWS EC2, Scikit-learn

2 October 2025

- Created an end-to-end YouTube sentiment analysis pipeline processing over 10,000 user comments, enhancing sentiment classification performance through NLP preprocessing techniques.
- Tracked multiple model experiments using MLflow and DVC, enabling reproducible training and systematic comparison of models built with scikit-learn and NLP libraries.
- Deployed the pipeline on AWS using Docker and exposed predictions via Flask REST APIs, facilitating scalable and reproducible inference.

Smart Product Pricing

Keras, Hugging Face Transformers, ResNet50, OpenCV
[Kaggle](#)

2 October 2025

- Developed an NLP and CV pipeline to analyze 150,000 image and text data using transformer-based text encoders and CNN-based image embeddings, integrating them through a fusion neural network for price prediction.
- Implemented data preprocessing techniques, including text cleaning, tokenization, and streaming image feature extraction with ResNet and CLIP representations to manage large datasets.
- Built and fine-tuned models using TensorFlow and scikit-learn, achieving a rank of 142 out of 50,000 participants.

Skills

- SQL
- Python
- C
- JavaScript
- HTML
- CSS
- Scala
- Machine Learning
- Deep Learning
- Computer Vision
- LLM Fine Tuning
- RAG
- LLMs
- prompt engineering
- Natural Language Processing (NLP)
- TensorFlow
- PyTorch
- scikit-learn
- Hugging Face
- Pandas
- NumPy
- Langchain
- Flask
- FastAPI
- Amazon Web Services (AWS)
- Streamlit
- ETL Pipelines
- Data Structures and Algorithms
- REST API
- Spark
- ML engineering
- software engineering with an ML focus
- MLOps tools
- PostgreSQL
- Vector Databases (Chroma)
- Docker
- MLflow
- DVC
- CI/CD
- Linux
- GCP
- Azure
- Kubernetes
- Airflow
- Kubeflow
- cloud platforms

Extracurricular

- Core Member - Entrepreneurship Cell, Jadavpur University
May 2024 – Present